

Accounting and Budgetary Systems for Engineers - INFO7225

MGEN

Introduction to Managerial Accounting

Professor Shiaoming Shi
College of Engineering
Northeastern University



Chapter 9

Accounting Information Systems and Business Processes: Part II

Accounting Subsystem.

Because the cost of goods sold is likely to be the largest expense on a manufacturing firm's income statement, a critical part of the production process is an AIS's cost accounting subsystem. The cost accounting subsystem provides important control information (e.g., variance reports reflecting differences between actual and standard production costs) and varies with the size of the company and the types of product produced. As you might guess, a bakery producing baked goods would collect very different data in its AIS compared to an automobile manufacturer. Cost accounting subsystems for manufacturing organizations are commonly job costing, process costing, or activity-based costing systems.

A **job costing information system** keeps track of the specific costs for raw materials, labor, and overhead associated with each product or group of products, called a “job.” This type of costing system is most appropriate for manufacturers of large-scale or custom products, such as home builders or book publishers. Manufacturers of homogeneous products (such as soft drinks or toothbrushes) that are produced on a regular and continuous basis use a **process costing information system**. In this system, it is not feasible or practical to keep track of costs for each item or group of items produced. Instead, process costing systems use averages to calculate the costs associated with goods in process and finished goods produced.

Activity-based costing systems help managers describe processes, identify cost drivers of each process, and then determine the unit costs of products created in each process. By studying their business processes, managers are in a better position to recognize opportunities to improve those processes. Thus, activity-based costing gives managers a better understanding of their processes, an improved ability to allocate indirect costs to those processes, and a better understanding of the true cost of each product. The systems flowchart in **Figure 9-7** shows a

LEARNING OBJECTIVES

After completing this segment, you should be able to

- Distinguish between financial and managerial accounting
- Identify the major items in the income statement of various types of firms
- Determine and recognize the fixed and variable components of costs
- Understand the basics of cost accounting in a manufacturing company
- Describe and Identify the three major components of product costs

THREE PRIMARY RESPONSIBILITIES OF MANAGEMENT

- Involves setting goals and forming the plans to achieve those goals.
- Financial (e.g., budgeting) and non-financial

- Involves the day-to-day activities and the monitoring of the planning objectives that were put into place.
- Purpose: help in planning functions and to facilitate coordination within the organization

- Determines whether plans are being followed and whether progress is being made as planned toward the fulfillment of organizational goals and objectives.
- Also involves taking corrective measures in case of deviations identified in the course of action.

Financial and Managerial Accounting Comparative

COMMUNICATION THROUGH REPORTING	FINANCIAL ACCOUNTING	MANAGERIAL ACCOUNTING
Users of reports	External users: stockholders, creditors, regulators	Internal users: managers, officers, and other employees
Types of reports	Financial statements: balance sheet, income statement, cash-flow statement, etc.	Internal reports: job cost sheet, cost of goods manufactured, production cost report, etc.
Frequency of reports	Quarterly; annually	As frequently as needed
Purpose of reports	Helps those external users make decisions: credit terms, investment, and other decisions	Assists the internal users in the planning and control decision-making process
Focus of reports	Pertains to company as a whole Uses GAAP structure Composed from a multitude or combination of other more individual data	Pertains to departments, sections of the business Very detailed reporting No GAAP constraints
Nature of reports	Monetary	Monetary and nonmonetary information
Verification of reports	Audited by CPA	No independent audits



Costs Incurred for Different Types of Business

Costs Incurred for Different Types of Business

Type of Business	Costs Incurred
_____ Business	• Direct labor • Plant and equipment • Manufacturing overhead • Raw materials
_____ Business	• Lease on retail space • Merchandise inventory • Retail sales staff
_____ Business	• Billing and collections • Computer network equipment • Professional staff

Table 2.1 Some costs, such as raw materials, are unique to a particular type of business. Other costs, such as billing and collections, are common to most businesses, regardless of the type.

WELCH & GRAHAM, ATTORNEYS AT LAW
Income Statement
For the Year Ended December 31, 2017

Service Revenue		\$1,500,000
Operating Expenses		
Administrative Salaries	\$ 150,000	
Attorney Salaries	750,000	
Office Expenses	12,000	
Office Rent	20,000	
Paralegal Salaries	100,000	
Office Utilities Expenses	<u>12,500</u>	
Total Operating Expenses		<u>1,044,500</u>
Operating Income		<u><u>\$ 455,500</u></u>

WHICHARD & KLEIN, LLP
Income Statement
For the Year Ended December 31, 2019

Service Revenue	\$412,000
Operating Expenses	
Salaries	210,000
Office Expense	35,000
Office Equipment	9,000
Administrative Salaries	45,000
Utilities	11,000
Miscellaneous	7,500
Total Operating Expenses	<u>317,500</u>
Operating Income	<u><u>\$ 94,500</u></u>

KOELLER MANUFACTURING
Income Statement
For the Month Ended March 31, 2017

Sales	\$214,000
– Cost of Goods Sold	<u>102,000</u>
Gross Profit	112,000
– Operating Expenses	<u>80,000</u>
Operating Income	<u><u>\$ 32,000</u></u>

Chapter 2. EA1

Magio Company manufactures kitchen equipment used in hospitals. They distribute their products directly to the customer and, for the year ending 2019, they reported the following revenues and expenses. Use this information to construct an income statement for the year 2019.

Sales revenue	\$985,000
Cost of goods sold	489,000
Operating expenses	245,000

Chapter 2. EA2

Park and West, LLC, provides consulting services to retail merchandisers in the Midwest. In 2019, they generated \$720,000 in service revenue. Their total cost (fixed and variable) per client was \$2,500 and they served 115 clients during the year. If operating expenses for the year were \$302,000 what was their net income?

Park and West, LLC	
Income Statement for Year Ended 2019	

Chapter 2. EA3

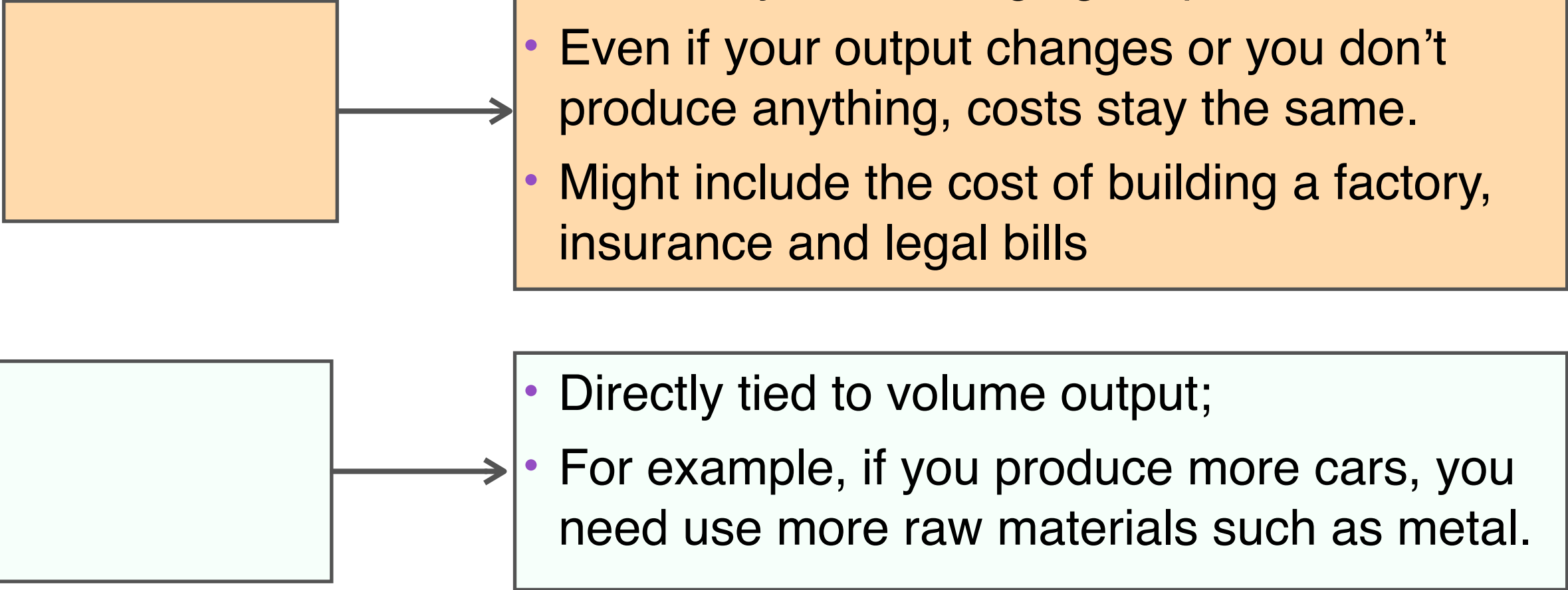
Canine Couture is a specialty dog clothing boutique that sells clothing and clothing accessories for dogs. In 2019, they had gross revenue from sales totaling \$86,500. Their operating expenses for this same period were \$27,500. If their Cost of Goods Sold (COGS) was 24% of gross revenue, what was their net operating income for the year?



Identify Basic Cost Behavior Patterns

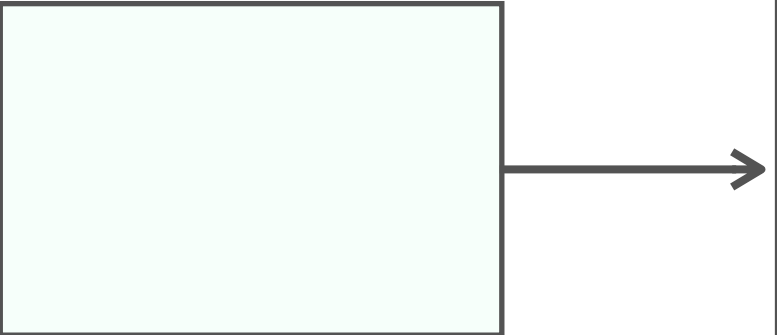
- 
- 
- Associated with the acquisition or production of goods and products;
 - When products are purchased for resale, the cost of goods is recorded as an _____ on the company's _____. It is not until the products are sold that they become an _____ on the _____;
 - By moving product costs to the expense account for the cost of goods sold, they are easily matched to the sales revenue income account.

- 
- 
- Typically related to a particular time period instead of attached to the production of an asset;
 - Treated as an _____ in the period incurred (examples include many sales and administrative expenses).

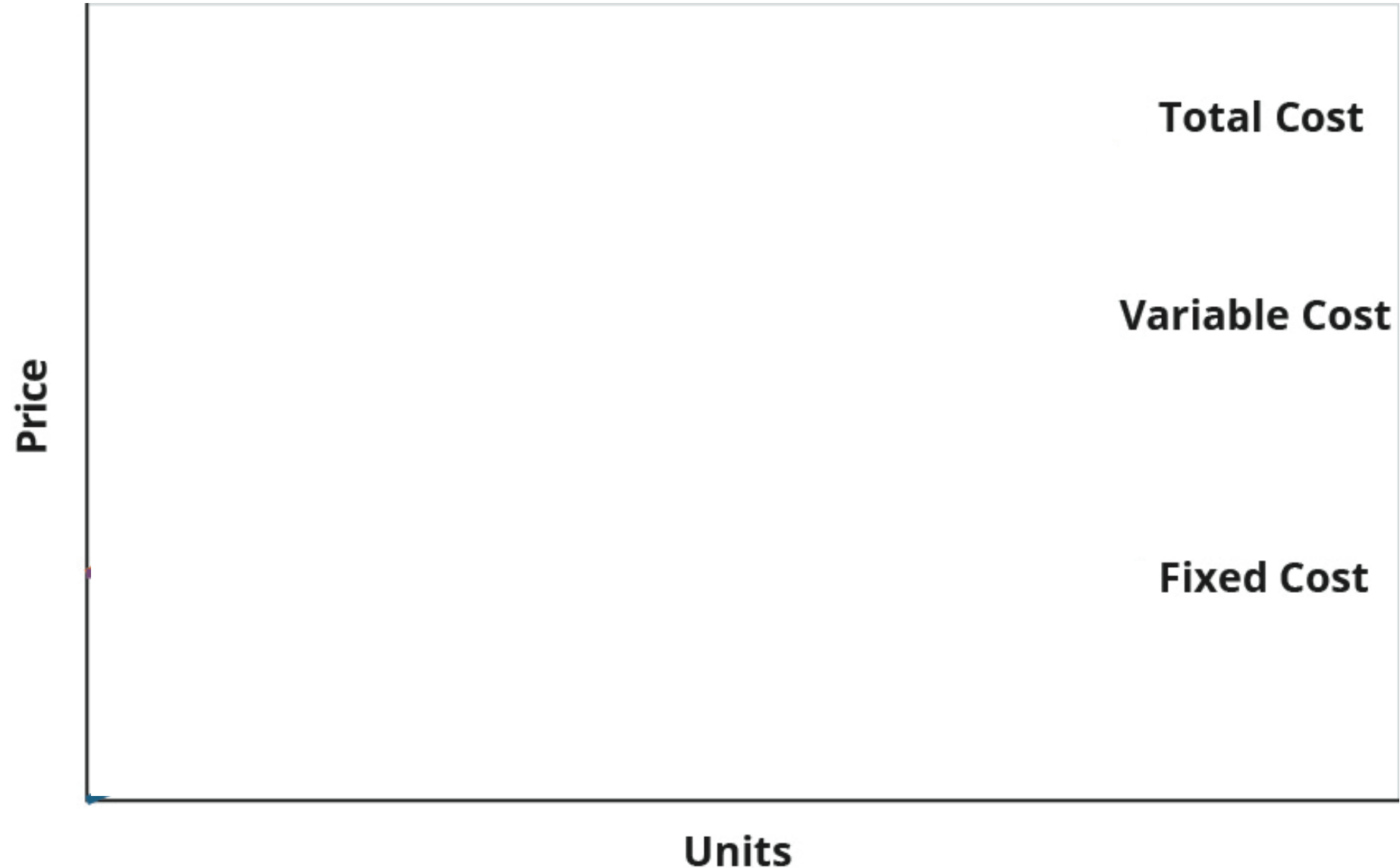
- 
- Don't vary with changing output.
 - Even if your output changes or you don't produce anything, costs stay the same.
 - Might include the cost of building a factory, insurance and legal bills

- Directly tied to volume output;
- For example, if you produce more cars, you need use more raw materials such as metal.

- 
- Labor might be an example of such cost.
 - If you produce more cars, you need to employ more workers; this is a variable cost.
 - However, even if you didn't produce any cars, you may still need some workers to look after an empty factory.

- 
- The cost of producing an extra unit.
 - If the total cost of 3 units is \$1550, and the total cost of 4 units is \$1900. The ~ cost of the 4th unit =

Total Cost as the Sum of Total Fixed Costs and Total Variable Costs



Tony's T-Shirts Cost Options

	Cost	Type of Cost ?	Relevant Range
Lease on Screen-Printing Machine	\$2,000 per month		0–2,000 T-shirts per month
Employee	\$10 per hour		20 shirts per hour
Tony's Salary	\$2,500 per month		N/A
Screen-Printing Ink	\$0.25 per shirt		N/A
Building Rent	\$1,500 per month		2 screen-printing machines and 2 employees

Functions of Cost Equations

Cost Information for Eagle Electronics

Cost Incurred	Fixed or Variable ?	Cost
Lease on manufacturing equipment		\$50,000 per year
Supervisor salary		\$75,000 per year
Direct materials		\$50 per unit
Direct labor		\$20 per unit

1. Determine total fixed costs: $\$ \underline{\hspace{2cm}} + \$ \underline{\hspace{2cm}} = \$ \underline{\hspace{2cm}}$

2. Determine variable costs per unit: $\$ \underline{\hspace{1cm}} + \$ \underline{\hspace{1cm}} = \$ \underline{\hspace{1cm}}$

3. Complete the cost equation (# of units = x): $Y = \$ \underline{\hspace{2cm}} + \$ \underline{\hspace{1cm}}x$

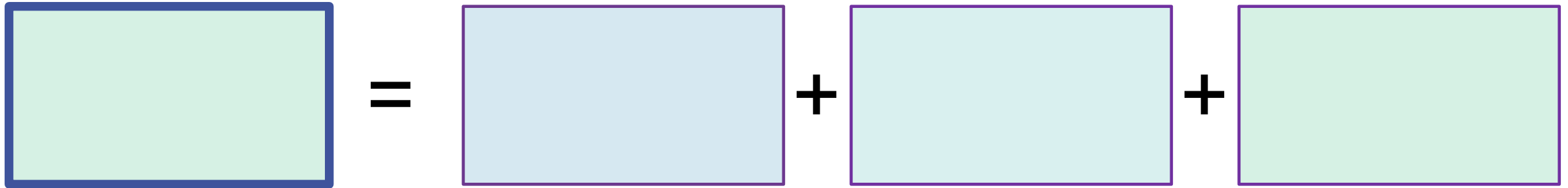
Chapter 2. EA8

Suppose that a company has fixed costs of \$18 per unit and variable costs \$9 per unit when 15,000 units are produced. What are the fixed costs per unit when 12,000 units are produced?



Manufacturing Costs (Aka Product Costs)

Manufacturing Costs (also called Product Costs)



KEY TERMS

Direct labor

Labor directly related to the manufacturing of the product or the production of a service

Direct materials

Materials used in the manufacturing process that can be traced directly to the product

Manufacturing overhead

Costs incurred in the production process that are not economically feasible to measure as direct material or direct labor costs; examples include indirect material, indirect labor, utilities, and depreciation

Manufacturing Overhead

- Costs that support production but are not direct materials or direct labor are considered overhead.
- Manufacturing overhead has three components:
 - 1) Indirect materials**

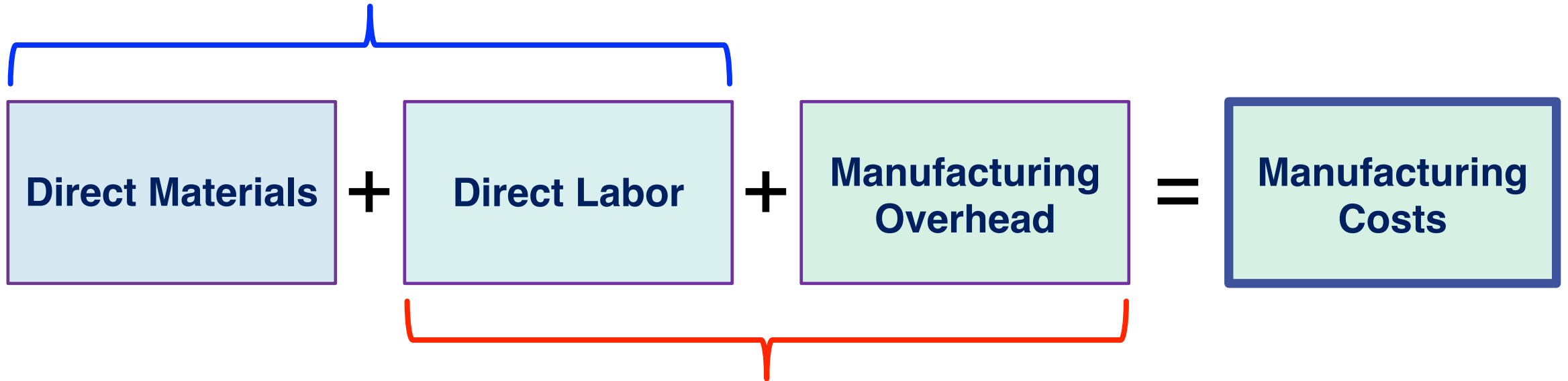
Materials used in production but not traced to specific products because the net informational value from the time and effort to trace the cost to each individual product produced is impossible or inefficient.
 - 2) Indirect labor**

Labor costs of those employees associated with the manufacturing process, but whose contributions are not directly traceable to the final product, and
 - 3) Overhead**

Costs that are necessary for production but not efficient to assign to individual product production. Examples of typical overhead costs are insurance, production facility electricity, warehouse rent, and depreciation of equipment.

Manufacturing Costs (also called Product Costs)

 Costs



 Costs

KEY TERMS

Prime costs

direct material expenses and direct labor costs

Conversion costs

Total of labor and overhead for a product; the costs that “convert” the direct material into the finished product

Period costs

Typically related to a particular time period instead of attached to the production of an asset; treated as an expense in the period incurred (examples include many sales and administrative expenses)

Chapter 2. EA 5. Rose Company has a relevant range of production between 10,000 and 25,000 units. The following cost data represents average cost per unit for 15,000 units of production.

	Average Cost per Unit
Direct materials	\$12
Direct labor	10
Indirect materials	2
Fixed manufacturing overhead	4
Variable manufacturing overhead	3
Fixed selling and administrative expenses	8
Variable sales commissions	25

- A. If 10,000 units are produced, what is the variable cost per unit?
- B. If 18,000 units are produced, what is the variable cost per unit?
- C. If 21,000 units are produced, what are the total variable costs?
- D. If 11,000 units are produced, what are the total variable costs?

Chapter 2. EA 5. Rose Company has a relevant range of production between 10,000 and 25,000 units. The following cost data represents average cost per unit for 15,000 units of production.

	Average Cost per Unit
Direct materials	\$12
Direct labor	10
Indirect materials	2
Fixed manufacturing overhead	4
Variable manufacturing overhead	3
Fixed selling and administrative expenses	8
Variable sales commissions	25

- E. If 19,000 (or 23,000) units are produced, what are the total manufacturing overhead costs incurred?
- F. If 19,000 (or 25,000) units are produced, what are the per unit manufacturing overhead costs incurred?



Contribution Margin and Break-Even Analysis



Contribution Margin

Example 1. You are evaluating orders from two new customers, but you will only be able to accept one of the orders without increasing your fixed costs. Management has directed you to choose the one that is most profitable for the company. Customer A is ordering 500 units and is willing to pay \$200 per unit, and these units have a contribution margin of \$60 per unit. Customer B is ordering 1,000 units and is willing to pay \$140 per unit, and these units have a contribution margin ratio of 40%. Which order do you select and why?



Break-even Analysis



Break-even Analysis

Example 2. Cindy Lou's, Inc. produces and sells blankets for dorm room beds and lofts. The selling price of each blanket is \$80 and the variable costs per blanket are \$48. Fixed costs are \$4,000.

- A. What is the break-even point in units and dollars?
- B. Present the contribution margin income statement at break-even.
- C. If Cindy Lou sells 150 blankets, what will be her income?
- D. If Cindy Lou has a contribution margin of \$6,400, how many blankets did she sell?
- E. If Cindy Lou's fixed costs change to \$4,640, how many additional units will Cindy need to sell to break even?

Example 2. Cindy Lou's, Inc. produces and sells blankets for dorm room beds and lofts. The selling price of each blanket is \$80 and the variable costs per blanket are \$48. Fixed costs are \$4,000.

A. What is the break-even point in units and dollars?

Example 2. Cindy Lou's, Inc. produces and sells blankets for dorm room beds and lofts. The selling price of each blanket is \$80 and the variable costs per blanket are \$48. Fixed costs are \$4,000.

B. Present the contribution margin income statement at break-even.

Example 2. Cindy Lou's, Inc. produces and sells blankets for dorm room beds and lofts. The selling price of each blanket is \$80 and the variable costs per blanket are \$48. Fixed costs are \$4,000.

C. If Cindy Lou sells 150 blankets, what will be her income?

Example 2. Cindy Lou's, Inc. produces and sells blankets for dorm room beds and lofts. The selling price of each blanket is \$80 and the variable costs per blanket are \$48. Fixed costs are \$4,000.

D. If Cindy Lou has a contribution margin of \$6,400, how many blankets did she sell?

Example 2. Cindy Lou's, Inc. produces and sells blankets for dorm room beds and lofts. The selling price of each blanket is \$80 and the variable costs per blanket are \$48. Fixed costs are \$4,000.

E. If Cindy Lou's fixed costs change to \$4,640, how many additional units will Cindy need to sell to break even?



Classification of Manufacturing Costs

Examples of Classifications of Manufacturing Costs (Ch2.2; Table 2.8)

Cost	Classification	Fixed or Variable
Wages of production employees		
Production supervisor salary		
Raw materials used in production		
Glue and adhesives		
Straight-line depreciation on factory equipment		

Chapter 2. EA 4. LO 2.2

Hicks Contracting collects and analyzes cost data in order to track the cost of installing decks on new home construction jobs. The following are some of the costs that they incur. Classify these costs as fixed or variable costs and as product or period costs.

F V Pro Per

- Lumber used to construct decks (\$12.00 per square foot)
- Carpenter labor used to construct decks (\$15 per hour)
- Construction supervisor salary (\$45,000 per year)
- Depreciation on tools and equipment (\$6,000 per year)
- Selling and administrative expenses (\$35,000 per year)
- Rent on corporate office space (\$34,000 per year)
- Nails, glue, and other materials required to construct deck (varies per job)

Chapter 4. EA 1. Little Things manufactures toys. For each item listed, identify whether it is a product cost, a period cost, or not an expense.

a product cost a period cost not an expense

- A. internet provider services
- B. material expense
- C. raw materials inventory
- D. production equipment rental
- E. showroom rental
- F. factory employee salary
- G. Human Resource Director salary

Chapter 4. EA 2

Expenses Involved in Lacrosse Stick Production

Lacrosse Stick Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Carbon, fiberglass			
Administrative building rent			
Accountant salary			

Chapter 4. EA 2

Expenses Involved in Lacrosse Stick Production

Lacrosse Stick Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Factory building depreciation			
Strings for the pocket			
Advertising			
Production supervisor salary			

Chapter 4. EA 2

Expenses Involved in Lacrosse Stick Production

Lacrosse Stick Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Paint for sticks			
Research and development costs			
Wages of person who strings the sticks			

Chapter 4. EA 2

Expenses Involved in Lacrosse Stick Production

Lacrosse Stick Production Costs	Manufacturing or Sales & Administration Cost?	If Manufacturing: Direct Materials, Direct Labor, or Overhead?	If Overhead: Indirect Materials, Indirect Labor, or Other?
Cutting machine depreciation			
Human resources salaries			
Factory maintenance			

Chapter 2. PA 2. Listed as follows are various costs found in businesses. Classify each cost as a fixed or variable cost, and as a product and/or period cost.

		FC	VC	Product Cost	Period Cost
A	Wages of administrative staff				
B	Shipping costs on merchandise sold				
C	Wages of workers assembling computers				
D	Cost of lease on factory equipment				
E	Insurance on factory				
F	Direct materials used in production of lamps				
G	Supervisor salary, factory				
H	Advertising costs				
I	Property taxes, factory				
J	Health insurance cost for company executives				
K	Rent on factory				

Chapter 4. EA 3

EA 3. [LO 4.2](#) Burnham Industries incurs these costs for the month:

Direct materials	\$2,000
Direct labor	3,000
Factory depreciation expense	3,500
Factory utilities expense	750
CEO's salary	4,000

- A. What is the prime cost?
- B. What is the conversion cost?

Chapter 2. PA 5. Wachowski Company reported these cost data for the year 2017. Use the data to compute the following costs.

1.	Factory maintenance costs	\$ 90,000
2.	Direct labor, wages	352,000
3.	Direct labor, health insurance	32,000
4.	Indirect labor, health insurance	15,000
5.	Health insurance for production supervisor	6,500
6.	Administrative costs	55,000
7.	Rental of office space for administrative staff	17,500
8.	Sales commissions	52,500
9.	Direct material	1,230,000
10.	Indirect materials	632,000
11.	Advertising expense	39,000
12.	Depreciation on factory building	62,000
13.	Indirect labor, wages	70,000
14.	Production supervisor's salary	32,000

A Total prime costs

B Total manufacturing overhead costs

C Total conversion costs

D Total product costs

E Total period costs

Chapter 4. EA4. Marzoni's records show raw materials inventory had a beginning balance of \$200 and an ending balance of \$300. If the cost of materials used during the month was \$900, what were the purchases made during the month?

Determining COGS for a Manufacturing Firm

Materials Inventory (beginning balance)
 + Net Material Purchases
 = Materials Available for Use
 – Materials Inventory (ending balance)
 = Direct Materials Used in Production

(Chapter 2; Fig. 2.8) **KOELLER MANUFACTURING**
Schedule of Cost of Goods Manufactured
For the Month Ended March 31, 2017

Work in Process Inventory (beginning balance)		\$ 75,000
Current Manufacturing Costs:		
Direct Material	\$15,000	
Direct Labor	25,000	
Manufacturing Overhead	<u>23,000</u>	
Total Manufacturing Costs		<u>63,000</u>
Total Cost of Work in Process		138,000
– Work in Process, ending balance		<u>43,000</u>
<u>Cost of Goods Manufactured</u>		<u>\$ 95,000</u>

(Chapter 2; Fig. 2.9) **KOELLER MANUFACTURING**
Schedule of Cost of Goods Sold
For the Month Ended March 31, 2017

Beginning Finished Goods Inventory	\$ 65,000
+ <u>Cost of Goods Manufactured</u>	<u>95,000</u>
Goods Available for Sale	160,000
– Ending Finished Goods Inventory	<u>58,000</u>
Cost of Goods Sold	<u>\$102,000</u>



Work-in-Process

WHAT IS A WORK-IN-PROGRESS (WIP)?

- The term work-in-progress (WIP) is a production and supply-chain management term describing partially finished goods awaiting completion.
- WIP refers to the raw materials, labor, and overhead costs incurred for products that are at various stages of the production process.
- WIP is a component of the inventory asset account on the balance sheet. These costs are subsequently transferred to the finished goods account and eventually to the cost of sales.

WHAT IS A WORK-IN-PROGRESS (WIP)?

- The WIP figure reflects only the value of those products in some intermediate production stages.
- This excludes the value of raw materials not yet incorporated into an item for sale.
- The WIP figure also excludes the value of finished products being held as inventory in anticipation of future sales.
- A WIP is different from a finished good which refers to a product that is ready to be sold to the consumer.

(<https://www.investopedia.com/terms/w/workinprogress.asp>)

COGS for a Merchandising Firm

Beginning Inv.

Cost of Goods

Ending Inventory

Cost of Goods

COGS for a Manufacturing Firm

Beginning Inv.

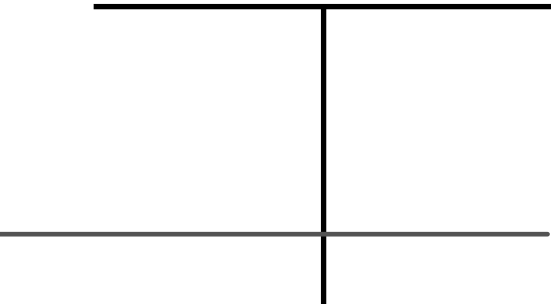
Cost of Goods

Ending Inventory

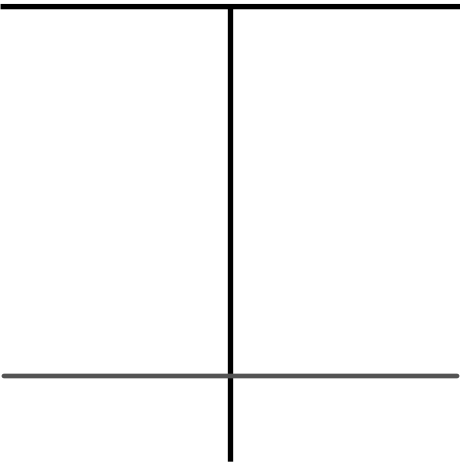
Cost of Goods

Flow of Materials from Raw Materials to Finished Goods

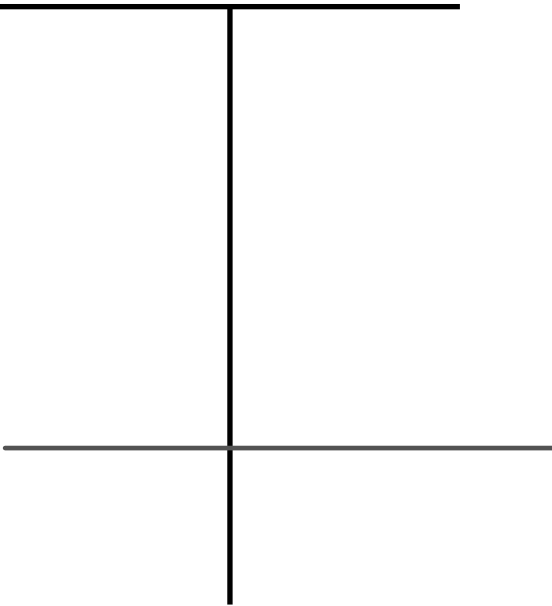
Materials Inventory



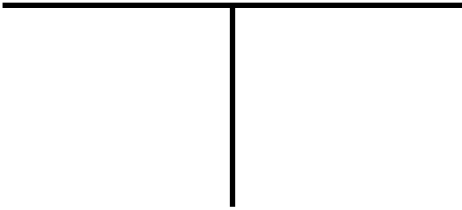
WIP Inventory



FG Inventory



COGS



Ch4, PB4. Summary information from a company's job cost sheets shows the following information:

Job	April	May	June	Completed	Sold
BB3	\$3,500	\$1,500		May	Not sold
BB4	9,231			April	May
BB5		2,540		May	June
BB6		3,230	\$1,434	July	Not sold

What are the balances in the work in process inventory, finished goods inventory, and cost of goods sold for April, May, and June?

Ch4, PB4. Summary information from a company’s job cost sheets shows the following information:

Job	April	May	June	Completed	Sold
BB3	\$3,500	\$1,500		May	Not sold
BB4	9,231			April	May
BB5		2,540		May	June
BB6		3,230	\$1,434	July	Not sold

What are the balances in the work in process inventory, finished goods inventory, and cost of goods sold for April, May, and June?

	April				May				June				July			
	BB3	BB4	BB5	BB6	BB3	BB4	BB5	BB6	BB3	BB4	BB5	BB6	BB3	BB4	BB5	BB6
WIP inventory																
	Total =				Total =				Total =							
FG inventory																
	Total =				Total =				Total =							
COGS																
	Total =				Total =				Total =							

Job	April	May	June	Completed	Sold
BB3	\$3,500	\$1,500		May	Not sold
BB4	9,231			April	May
BB5		2,540		May	June
BB6		3,230	\$1,434	July	Not sold

BB3	April		May		June		July	
WIP Inventory			WIP Inventory		WIP Inventory		WIP Inventory	
FG Inventory			FG Inventory		FG Inventory		FG Inventory	
COGS			COGS		COGS		COGS	

BB4	April		May		June		July	
WIP Inventory			WIP Inventory		WIP Inventory		WIP Inventory	
FG Inventory			FG Inventory		FG Inventory		FG Inventory	
COGS			COGS		COGS		COGS	

Job	April	May	June	Completed	Sold
BB3	\$3,500	\$1,500		May	Not sold
BB4	9,231			April	May
BB5		2,540		May	June
BB6		3,230	\$1,434	July	Not sold

BB5	April		May		June		July		BB6	April		May		June		July	
WIP Inventory			WIP Inventory		WIP Inventory		WIP Inventory		WIP Inventory			WIP Inventory		WIP Inventory		WIP Inventory	
FG Inventory			FG Inventory		FG Inventory		FG Inventory		FG Inventory			FG Inventory		FG Inventory		FG Inventory	
COGS			COGS		COGS		COGS		COGS			COGS		COGS		COGS	

-
- **Job order costing**
 - **Process costing**
 - **Activity-based costing**



Thank you!
Questions?